

Autonomous AI Shopping Assistant

Built using MachinaOS

Contents

1	Introduction	2
2	Project Goals	2
3	Key Features	2
4	System Architecture	3
4.1	Workflow Architecture	3
4.2	Core Components	3
5	Workflow Explanation	3
6	Memory System	4
7	Cross-Platform Product Comparison	4
8	Autonomous Deal Monitoring	5
9	Example Queries	5
10	Technology Stack	5
11	Challenges Faced	6
12	Future Improvements	6
13	Conclusion	6
14	GitHub Repository	6

1 Introduction

The Autonomous AI Shopping Assistant is an agentic AI system built using MachinaOS. The goal of the project is to create a persistent AI assistant capable of tracking products, monitoring prices, comparing products across shopping platforms, and automatically sending deal alerts through Telegram.

Unlike traditional chatbots that only respond to user prompts, this system continuously operates in the background using scheduling, memory, and live web interaction. The assistant remembers user preferences, performs product searches autonomously, and intelligently filters unreliable or low-quality listings.

The project explores the concept of agentic AI workflows where AI models orchestrate memory, scheduling, tool usage, and decision making instead of only generating conversational text.

2 Project Goals

The primary objectives of the project are:

- Build an autonomous shopping assistant using MachinaOS.
- Implement persistent memory for tracking products and preferences.
- Perform cross-platform product comparison.
- Automate deal monitoring using scheduled execution.
- Integrate Telegram as the primary user interface.
- Use live web scraping for real-time product information.
- Explore agentic AI system design.

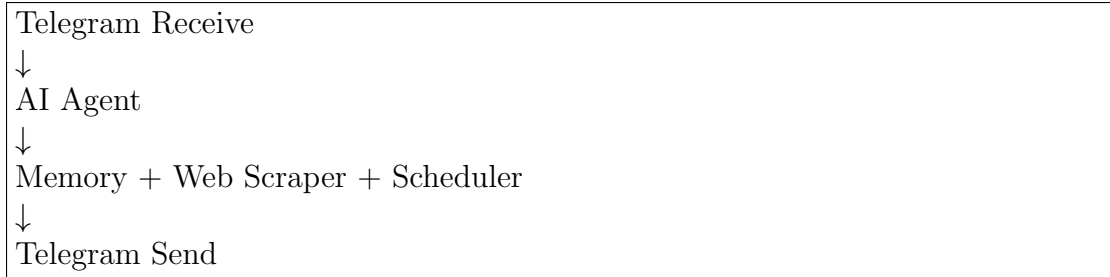
3 Key Features

- Autonomous deal monitoring
- Persistent shopping memory
- Telegram integration
- Cross-platform shopping search
- AI-based recommendation engine
- Product comparison system
- Scheduled background execution
- Live web scraping
- Personalized shopping preferences
- Intelligent product filtering

4 System Architecture

The project uses a lightweight agentic workflow architecture.

4.1 Workflow Architecture



4.2 Core Components

Component	Purpose
Telegram Receive	Receives user shopping requests and commands.
AI Agent	Main shopping intelligence engine responsible for reasoning and orchestration.
Memory	Stores tracked products, budgets, and user preferences.
Web Scraper	Extracts live product information from shopping platforms.
Cron Scheduler	Automatically triggers periodic deal scans.
Telegram Send	Sends recommendations and deal alerts back to users.

5 Workflow Explanation

1. The user sends a shopping request through Telegram.
2. The AI agent analyzes the request.
3. The memory system stores or recalls tracked products and preferences.
4. The AI dynamically generates shopping search URLs.
5. The web scraper extracts product information such as:
 - Product name
 - Price
 - Rating
 - Stock status
 - Product link
6. The AI compares products across stores.
7. Low-quality or suspicious listings are filtered.

8. The best recommendations are returned through Telegram.
9. The scheduler periodically triggers autonomous scans.

6 Memory System

The assistant uses persistent memory to behave like a long-term shopping companion rather than a stateless chatbot.

The memory stores:

- Tracked products
- Budget preferences
- Preferred stores
- Disliked brands
- Previous interactions

Example:

watch gaming mouse under 1500

The AI stores this preference and later performs autonomous scans using the scheduler.

7 Cross-Platform Product Comparison

The AI assistant dynamically searches multiple shopping platforms such as:

- Amazon
- Flipkart
- Croma
- Reliance Digital

The assistant compares:

- Price
- Ratings
- Seller reliability
- Product availability
- Value for money

The final recommendation is generated using AI-based reasoning.

8 Autonomous Deal Monitoring

One of the most important features of the project is autonomous execution.

The Cron Scheduler periodically activates the AI agent automatically. During scheduled runs, the assistant:

- Recalls tracked products from memory
- Searches shopping websites
- Compares prices and ratings
- Detects meaningful deals
- Sends Telegram alerts automatically

This transforms the system from a traditional chatbot into an autonomous AI assistant.

9 Example Queries

find wireless mouse under 1000

watch gaming keyboard under 2500

compare Logitech M331 vs HP Z3700

check deals

10 Technology Stack

- MachinaOS
- Gemini API
- Telegram Bot API
- Web Scraping Tools
- AI Memory System
- Cron Scheduler

11 Challenges Faced

During development, several practical challenges were encountered:

- Managing workflow complexity
- Handling memory persistence
- Designing autonomous execution logic
- Avoiding repeated notifications
- Building reliable web scraping logic
- Integrating multiple tools into a single AI workflow

These challenges helped improve understanding of agentic AI systems and orchestration workflows.

12 Future Improvements

Future upgrades planned for the project include:

- Price trend analytics
- Better personalization
- Auto-purchase approval flow
- Warranty reminders
- Multi-user support
- Advanced ranking algorithms
- Product lifecycle tracking

13 Conclusion

The Autonomous AI Shopping Assistant demonstrates how modern AI systems can move beyond traditional chatbots into persistent, autonomous, tool-using agents.

By combining memory, scheduling, web scraping, and AI reasoning, the system behaves more like a smart shopping companion capable of independently monitoring products and assisting users in real-world shopping tasks.

The project also highlights the power of MachinaOS for building agentic AI workflows involving memory, orchestration, autonomous execution, and live web interaction.

14 GitHub Repository

GitHub Repository Link:

<https://github.com/Siddharth-6/Autonomous-AI-Shopping-Assistant>